

Deepti Ahlawat

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LinkedIn

Education

University of Maryland, Baltimore County (UMBC)

Ph.D. in Computer Science

Baltimore, MD

Aug. 2026 - Present

New Jersey Institute of Technology

M.S. in Data Science

Newark, NJ

Sept. 2022 - May 2024

Maharshi Dayanand University

M.S. in Mathematics

Haryana, India

Aug. 2018 - Dec. 2020

Research Projects

- **Retrieval-Side Defenses against Adaptive Prompt-Injection Attacks in RAG Chatbots.** In Preparation.
- **Learning from Failure: Seeding Agent Memory with Past Mistakes for LLM-Based Penetration Testing.** In Preparation.

Work Experience

University of Maryland, Baltimore County (UMBC)

Graduate Assistant – Grader, CMSC 203: Discrete Structures

Baltimore, MD

Aug. 2026 - Present

- Grade assignments and provide feedback to students in Discrete Structures, supporting course instruction and student learning.

Velor Pathways

Machine Learning Engineer

USA

Jun. 2023 - May 2026

- Developed Retrieval-Augmented Generation (RAG) systems using OpenAI, LangChain, and FAISS, improving response accuracy by 37%.
- Built LLM-based internal chatbots using GPT-4, Pinecone, and LangChain, reducing customer service query time by 42%.
- Fine-tuned open-source models including LLaMA2 and Falcon using Hugging Face and NLTK, increasing model accuracy by 28%.
- Deployed AI/ML systems using Docker, Kubernetes, and FastAPI, automated ML workflows with MLflow and Airflow, and developed Splunk dashboards to monitor cloud data and system activity.

Internships

Office of Institutional Effectiveness, New Jersey Institute of Technology

Data Scientist Intern

Newark, NJ

May 2023 - May 2024

- **Topic Analysis on Data Queries:**
 - Employed natural language processing techniques to conduct topic analysis on data queries, enhancing information retrieval processes by identifying primary topics and requests.
 - Streamlined identification of important themes and trends within data searches using word cloud and n-gram cloud approaches, improving information extraction.
 - Applied advanced analytics techniques to data queries for subject analysis, allowing for more efficient prioritizing and classifying of information retrieval operations.
- **Strategic Evaluation and Visualization Framework for Research University Rankings Enhancement:**
 - Directed data analysis and classification for colleges, segregating them into R1, R2, and R3 based on integrated 2019–20 doctoral degree conferral data and Fall 2020 faculty counts from IPEDS (HR).
 - Utilized PCA to visually depict research and development profiles by integrating fiscal year 2020 R&D expenditures and research staff data obtained from NSF Survey data.
 - Assembled data from diverse sources including QS News, US News, World University Rankings, Carnegie Classification, and Washington Monthly. Employed R programming for data cleaning and preprocessing.
 - Transformed the consolidated dataset for visualization using MicroStrategy, crafting an extensive dashboard with variables spanning from 2020 to 2024.

Career Development Services, New Jersey Institute of Technology

Data Analytics Intern

Newark, NJ

Oct. 2022 - May 2023

- Independently crafted and optimized SQL queries, resulting in a 66% enhancement in data retrieval speed and operational efficiency.
- Engineered an automated data processing framework using script-based automation and mathematical formulas, reduc-

ing report generation time by 4 hours per instance.

- Applied methodologies to data cleaning and organization, leading to a 60% enhancement in data precision and dependability.
- Leveraged Power BI for data visualization, transforming complex datasets into intuitive visual representations to facilitate informed decision-making.

Sanskar Valley Public School

Mathematics Teacher

Haryana, India

Dec. 2020 - Aug. 2022

- Taught mathematics to high school students, covering algebra, geometry, calculus, and statistics.
- Organized and conducted extracurricular mathematics workshops and tutoring sessions.

Freelance

Private Tutor

Haryana, India

Jan. 2016 - Aug. 2022

- Provided one-on-one tutoring for high school students in mathematics, physics, and chemistry.
- Tailored tutoring sessions to address individual student challenges.
- Assisted students in preparing for competitive exams, focusing on problem-solving strategies and time management.

Skills

Languages & Frameworks	Python, PyTorch, TensorFlow, LangChain, OpenAI, Scikit-learn, FastAPI, Flask
Machine Learning & Gen AI	LLMs, RAG, Fine-tuning, Prompt Engineering, NLP, Recommendation Systems, Chatbots
Data & Storage	SQL, Pandas, NumPy, Spark, Hadoop, Hive, Delta Lake, Snowflake, BigQuery
MLOps & Deployment	Docker, Kubernetes, MLflow, SageMaker, Airflow, Git, CI/CD, FastAPI, REST APIs
Vector DBs & Tools	FAISS, Pinecone, Weaviate, ChromaDB, Elasticsearch, Milvus
Cloud Platforms	AWS (S3, EC2, Lambda, SageMaker), GCP, Azure
Visualization & Analysis	Matplotlib, Seaborn, Plotly, Tableau, Power BI

Academic Class Projects

- **Udemy: 100 Days of Code – Python Pro-Bootcamp**
 - Game Development – Blackjack, Snake Game, Pong Game, Build Own API, Fully Fledged Blog Website, Auto Job Applications on LinkedIn, Google App Store Analysis, Automate Birthday Emails/SMS, and Data Science with Google Trends.
- **Sentiment Analysis of Amazon Reviews Dataset**
 - Led initiatives to refine sentiment analysis for Amazon reviews.
 - Achieved a 15% increase in model accuracy and a 20% improvement in F1-score.
 - Resulted in more accurate and reliable insights for decision-making processes.
- **Data Mining Movie Recommendations**
 - Accomplished a 35% increase in client engagement and a 40% improvement in recommendation accuracy.
 - Created a more personalized user experience through effective data mining techniques.

Certifications

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| • DS101X: Statistical Thinking for Data Science and Analytics by ColumbiaX | <i>Aug. 2020</i> |
| • PH125.1X: Data Science: R Basics by Harvard | <i>Jul. 2020</i> |

Voluntary Memberships

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| • Graduate Student Association Member | <i>Sep. 2022 - May 2024</i> |
| • Graduate Women in Computing Science | <i>Sep. 2022 - May 2024</i> |